Check for updates

Received: 16 January 2021 | Accepted: 14 November 2022

DOI: 10.1111/poms.13963

POMS

ORIGINAL ARTICLE

Determining maximum shipping age requirements for shelf life and food waste management

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Handling Editor : Chelliah Sriskandarajah

Abstract

Products approaching the end of their shelf life on retail store shelves are more likely to result in food waste. For this reason, manufacturers establish shipping policies related to the age of the products that leave their warehouses, that is, they set a maximum age beyond which shipping to retail stores is no longer allowed. In practice, most existing policies are simple one-size-fits-all rules that do not accommodate the varying characteristics of the products. We offer a framework that manufacturers can use to determine *maximum shipping age thresholds* based on a Markov chain model, where the objective is to maximize profit, net of the cost of expiration at the warehouse and retail stores. We derive analytical insights about the impact of a shipping age threshold on food waste in the supply chain and obtain sufficient conditions under which a maximum shipping age threshold is suboptimal within the class of Ship-Oldest-First (a.k.a. First-In-First-Out) issuing policies. We also numerically investigate the relationship between different system parameters (e.g., demand rate, waste cost at warehouse and retail, warehouse inventory, total shelf life) and the optimal shipping age threshold. Using real data from our industry collaborator, we compute the optimal shipping age thresholds at the stock-keeping-unit (SKU) level for over 450 products and find that 9–10% of the SKUs currently have suboptimal shipping age thresholds. This presents an opportunity to improve profits by up to 8.7% and reduce food waste by up to 14.7%. These improvements correspond to up to $292,561 savings and 1846 truckloads of waste reduction annually. Our framework can be adopted by any firm and satisfies a much emphasized need in industry to control food waste through shelf life management.

KEYWORDS
consumer packaged goods, data-driven research, food waste, inventory issuance, Markov chain analysis, perishable inventory, practice-driven research

1 | INTRODUCTION

Food waste is a major problem of our times due to its negative implications for the environment and society. The United Nations estimates that as much as one-third of the food that is produced is never eaten. Organic waste which ends up in landfills generates greenhouse gases that are often compared in size to the emissions generated by China as a country (UN, 2013). Finally, food waste is a moral issue considering that 35.2 million people, including 5.3 million children, struggle with food insecurity in the United States alone (USDA, 2019).

One important driver of food waste is the aging of inventory in the supply chain. When products spend too much of their finite shelf life in the upstream stages of the supply chain and reach the retail store shelves with an advanced age, it increases the chance of expiration before sale to consumers, leading to retail-level food waste, as retailers commonly dispose of expired (or even soon-to-expire) foods from their store shelves. This is a major contributor to the $15B waste problem in the grocery industry (GMA, 2008). At the same time, the aging of products on store shelves may affect the perceived quality of products, which can impact consumer experience and, thus, future demand.

Causes of inventory aging in supply chains include operational inefficiencies (e.g., excess stock, unsynchronized

Accepted by Chelliah Sriskandarajah, after two revisions.

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product launches across retail chains) but also practices that offer operational benefits such as production batching or inventory issuing policies that reduce labor costs (Akkaş & Honhon, 2022). As a result, the issue is prevalent and can happen at substantial levels. For example, an audit study conducted at one of the warehouses of our industry collaborator, a large consumer-packaged goods (CPG) firm, revealed that the remaining shelf life (measured as the time until expiration as a percentage of total shelf life) of the inventory selected for shipments to retailers was 72% on average. Further, 1.02% of it had a remaining shelf life less than the minimum 6 weeks required by corporate headquarters for shipment to retail stores. As another example, a retailer we interviewed revealed that its stores occasionally receive products with only 1 month of remaining shelf life after those products spend 8 months in the manufacturers' distribution network. Such instances frustrate retailers and cause conflicts between trading partners. For this reason, manufacturers and retailers often establish rules regarding the age of inventory at reception at the retail stores; for example, they may impose a minimum number of remaining shelf life days (e.g., 30 days), below which the products may be turned down by the retailer. We have learned from several U.S. retailers that common practice is to require a minimum shelf life remaining of 70% for fresh products (e.g., dairy, packaged meat, and eggs) and 80% minimum for shelf stable products (e.g., canned goods, cereal boxes, etc.). A European distributor we spoke to practices the so-called "one-third – one-third – one-third rule," which states that a maximum of one-third of a product's shelf life should be spent at the manufacturer, no more than another one-third should take place at the distribution center, so that the product arrives with at least one-third of its remaining shelf life at the retail store. While the specific rules vary across countries and across supply chain partners, a common feature is that they are agnostic to product characteristics, such as the demand rate. Consequently, they may not be effective in terms of minimizing product waste: while a minimum 30-day remaining shelf life rule may be sufficient to avoid product expiration for a fast-selling product, it may result in significant expiration for a slow-moving SKU within the same product category.

Our study proposes a framework to determine shipping age thresholds that addresses the shortcomings of the existing industry rules by incorporating important factors such as product demand rate and waste cost. Note that, in practice, most rules are stated in terms of a *minimum remaining shelf life threshold* (i.e., time until expiration, as stated by the date label) but, in our model, we find it clearer to present the rules in terms of a *maximum shipping age threshold* (i.e., time since manufacturing date). Both approaches are equivalent because the shipping age of a product is equal to its total shelf life (i.e., time from manufacturing to expiration date) minus its remaining shelf life.

In practice, CPG manufacturers in the United States typically reimburse retailers, partially or fully, for the cost of products that expire at retail stores (GMA-AMR, 2008; GMA, 2008), which we refer to as the *waste cost*. For this reason, we frame the problem as an optimization decision from the perspective of a manufacturer selling products to multiple retailers. Also, in practice, the waste cost is typically higher for products that expire on retail shelves than at the warehouse because the former includes additional handling and logistics costs. Hence, from a profit-maximization perspective, manufacturers must think twice before shipping units to retail stores that are nearing the end of their shelf life.

We aim to answer the following questions:

(1) How should a manufacturer set shelf life management rules for their products shipped to retail stores? In particular, how should they set maximum shipping age thresholds beyond which units can no longer be shipped to retail stores?
(2) What are the trade-offs involved in setting these thresholds and what is the impact on product waste, both at the manufacturer and retailer levels?
(3) What are the benefits of optimizing the thresholds at the SKU level, by incorporating product characteristics such as the demand rate or waste cost?

In our analysis, we assume that the manufacturer follows a Ship-Oldest-First (SOF) policy for issuing inventory from its warehouse with a *shipping age threshold*, that is, inventory units are selected for shipment in decreasing order of age (oldest to youngest) but any inventory unit above the age threshold cannot be shipped to the retailer, that is, it is deemed "too old" and must be disposed of.[1] The manufacturer optimizes the shipping age threshold to maximize profit, which is equal to the gross profit from sales minus the waste costs at the warehouse and the retail stores.

We find that an older shipping age threshold is associated with less waste at the warehouse but more waste at the retailer, because the units which are received are on average older, leaving fewer time periods for consumers to select them. Thus, when setting the shipping age threshold, the manufacturer faces a trade-off between food waste at the warehouse and at retail stores. Further, we find that an older shipping age threshold is associated with less **total waste** in the supply chain, so that total supply chain waste is minimized when a pure SOF issuing policy is used, that is, no shipping age threshold is set. Yet, because the waste cost at the warehouse is lower than that at the retail stores, we show that profits can be maximized with a shipping age threshold that is strictly less than the product's shelf life; specifically, this occurs when the warehouse waste cost is less than the difference between the retail waste cost and the product's gross margin. Further, we show that, under mild conditions that are satisfied numerically, the profit function is unimodal in the shipping age threshold, that is, exhibits a unique peak. Finally, we numerically show that the optimal shipping age threshold decreases in the waste cost at the retail but increases in the waste cost at the warehouse as well as in the product's gross margin, demand rate, and shelf life.

We use a dataset from our industry collaborator, which consists of demand, cost, shelf life, and case size data on

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455 SKUs to determine the optimal shipping age threshold for each SKU separately. We compare the profit generated using our SKU-specific optimized thresholds with the current company practice of not shipping any unit that has less than 6 weeks of remaining shelf life: our method offers up to an 8.7% increase in profits and 14.7% reduction in total waste for SKUs that currently have suboptimal thresholds, which corresponds to up to $292,561 savings and 1846 truckloads of waste reduction annually. As a second benchmark to existing practice, we calculate total-waste-minimizing shipping age thresholds: we find that such thresholds increase profits by up to 6.8% and reduce the total waste by up to 16.1%. Thus, we propose two different approaches: a profit maximization approach and a total waste minimization approach. We show that both approaches significantly improve on current supply chain performance.

1.1 | Literature review and contribution

Our research contributes to the growing literature on food waste by addressing shelf life management, an important factor determining product expiration. For date-labeled products, *shelf life* is the time period between the manufacturing date and the expiration date. The latter appears on the date label affixed to the product and can be stated in numerous ways such as "best by,""enjoy by," "sell by," etc. For fresh products without date labels, *shelf life* is the time between the harvest date and when it looks visually unappealing to consumers.

Food waste has been studied through several lenses in the operations management literature that includes waste diversion (Ata et al., 2012, 2019; Lee & Tongarlak, 2017), package size and promotions (Kirci et al., 2018; Wu & Honhon, 2023), shelf space management (Akkaş, 2019; Atan Z.& Pan, 2020), retail revenue models (Belavina et al., 2016), store networks (Belavina, 2021), sales incentives (Akkaşs & Sahoo, 2020), and perishable inventory management (Akkaş & Honhon, 2022). Our study builds on this literature by addressing food waste through setting proper shelf life management rules upstream in the supply chain.

A recent industry study of *unsaleables* (the industry term used to refer to expired, damaged, or discontinued CPGs that are removed from retail shelves) includes "shelf life management" as one of the two collaborative best practice categories that are key to reducing unsaleables, arguing that manufacturers should *"factor in shelf life at time of sale—i.e., set a practical minimum shelf life for all customers"* as one of the top priority initiatives. Our paper responds to this call as we investigate inventory issuing rules based on shipping age thresholds for manufacturers sending products to retail stores. These rules help reduce waste in three ways. First, they reduce expiration at retail stores by preventing old inventory from making its way to the retail shelves. Second, they reduce the amount of waste going to landfills, because expired or *short-coded* (the typical term used in industry for products with little remaining shelf life) products are more likely to be donated in the upstream stages of the supply chain compared to the retail stores (a recent survey finds that 95% of manufacturers donate or recycle expired or short-coded food compared to only 42% of retailers; CHEP, 2014). Third, maximum shipping age threshold rules increase product freshness at retail stores, which reduces the chances of waste at the household level. Thus, although maximum shipping age threshold rules generate more waste at the upstream stages of the supply chain because products are disposed of and never shipped to retail stores), they may reduce the overall amount of product sent to landfills.

In the academic literature, shelf life management has largely been tackled from the perspective of shelf life extending treatments or biological tools (Ashie et al., 1996; Deegan et al., 2006; Lambert et al., 1991; Ma et al., 2017; Yousuf et al., 2018). For packaged fresh food, smart packaging solutions intend to extend shelf life by utilizing sensors that dynamically display expiry dates based on the gases released inside the package (Biji et al., 2015). Li et al. (2017) and Tromp et al. (2012) analyze the value of such packaging systems from an operations point of view, considering consumers' inventory issuing behavior, stock-outs, waste, and profits.

Hertog et al. (2014) and Van Elzakker et al. (2014) develop models that use remaining shelf life information in operational decision-making. A number of studies focus on predicting remaining shelf life in the cold chain by using temperature, humidity, or other factors that impact food quality with the help of traceability technology (Grunow & Piramuthu, 2013; Raab et al., 2008). In these studies, minimum shelf life information is typically considered as an input to other operational decisions, rather than a decision variable, as it is in our paper. The only studies we are aware of that treat remaining shelf life as a decision are Mohamadi et al. (2021) and Santos et al. (2022). Mohamadi et al. (2021) tackle contracting and supply chain coordination using remaining shelf life requirements. Our study differs from theirs in multiple ways: (i) we analyze a centralized supply chain, (ii) we tackle a tactical problem about inventory issuing rules related to aged inventory, as opposed to a strategic level contracting problem, (iii) we develop a perishable inventory model, in contrast to the game-theoretic model used in their analysis, and (iv) our study utilizes real-world data. On the other hand, Santos et al. (2022) use a mixed-integer linear optimization model to optimize what they refer to as the *minimum life upon receipt* in a one-stage and two-stage supply chain. Their study differs from ours in the following ways: (i) they do not use a real-life dataset, (ii) their analysis is purely numerical, and (iii) they do not consider a model in which the manufacturer is responsible for the cost of waste both at the manufacturer's warehouse and retail stores and, in fact, do not include waste costs in their profit function, which leads to very different trade-offs.

Most related to our paper, Akkaş and Honhon (2022) analyze the impact of a manufacturer's inventory replenishment and issuing policies on profits, food waste, and product freshness in a two-stage centralized setting with one manufacturer

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and one retailer. The study concludes that the optimal issuing policy is complex; the SOF policy is not always the profit-maximizing issuing policy, yet it minimizes the total waste in the supply chain. In contrast, we model a more complex supply chain with one manufacturer and multiple retailers, assume an order-up-to-replenishment policy, and focus on a specific issuing policy: SOF with a shipping age threshold. This issuing policy is not considered by Akkaş and Honhon (2022), who focus on the structure of the optimal replenishment and issuing policy. In contrast, the focus of our analysis is on properties of the optimal shipping age threshold and its impact on the manufacturer's profit and waste. To the best of our knowledge, our paper is the first study that optimizes shipping age thresholds in a two-stage supply chain.

The rest of the paper is organized as follows. Section 2 describes the setting at our industry collaborator and describes current practice regarding minimum shelf life requirements; Section 3 presents our model, the solution procedure, and analytical findings; Section 4 presents our numerical study; Section 5 presents the application of our framework at our collaborator; Section 6 discusses the implications of our research for practice; and finally, Section 7 concludes our paper. In an effort to target practitioners, we have intentionally kept the core of our paper light on mathematical concepts; the technical details of our analysis are confined to Section 3 and the Supporting Information Appendix A and B.

2 | SETTING

Our industry collaborator, which we refer to as *BevCo*, is a multibillion dollar food and beverage manufacturer based in the U.S. BevCo delivers its products directly to retail stores from its 350+ warehouses through the *Direct-Store-Delivery* (DSD) channel. DSD is a sales and delivery method under which the supplier's sales representatives generate replenishment orders on-site while performing other servicing tasks at the retail store (e.g., set up displays, reset planograms, update price tags, etc.). Under the DSD model, the manufacturer delivers the product directly to the retail store, bypassing the retailer's distribution center. In the United States, the DSD channel represents 24% of the sales volume at grocery stores (GMA-AMR, 2008). As is common with DSD suppliers, BevCo fully reimburses retailers for the waste cost incurred at their retail stores and picks up expired items after product delivery. A product is considered to be wasted once it reaches what BevCo refers to as its *best taste limit* date, which is printed on the package.

BevCo has established rules for the management of its products' shelf life that its warehouse managers are instructed to follow; these consist of three types of thresholds, termed by BevCo as follows:

- *Ship-to-satellite* threshold: the minimum remaining shelf life required for shipments from manufacturing plants to warehouses.
- *Ship-to-trade* threshold: the minimum remaining shelf life required for shipments from warehouses to retail stores.
- *Target-to-trade* threshold: the ideal remaining shelf life when the product arrives at the retail stores, which serves as a level that the supply chain should aim for.

These thresholds are set by BevCo based on the products' full shelf life, which is defined as the period between the time of manufacturing and the "best taste limit" date printed on the package. For example, all SKUs with a full shelf life of 25 weeks have their *target-to-trade* set at 19 weeks, *ship-to-satellite* at 16 weeks, and *ship-to-trade* at 6 weeks. As seen in Figure 1, the *target-to-trade* thresholds set by BevCo increase with the full shelf life of the products (on average, they correspond to about 76% of the full shelf life), but the other two thresholds remain constant past a certain level: for SKUs with a full shelf life of more than 12 weeks, the *ship-to-trade* threshold is set at 6 weeks and for products with more than 16 weeks of full shelf life, the *ship-to-satellite* is set at 16 weeks.[2] From our conversations with BevCo managers, it appears that these thresholds were established in an ad hoc fashion, rather than being the product of advanced optimized data analytics.

If a product's remaining shelf life reaches its *ship-to-satellite* threshold at the manufacturing plant, it is supposed to be written off and disposed of, that is, it cannot be shipped to a warehouse. Similarly, if a product's remaining shelf life reaches its *ship-to-trade* threshold at a warehouse, it is supposed to be written off and disposed of, that is, it cannot be shipped to a retail store. Imposing minimum shelf life rules has several benefits for BevCo. First, it forces manufacturing plant and warehouse managers to reduce inefficiencies, partially addressing the problem at its source. Second, a longer remaining shelf life for products on the store shelves can make products more attractive to consumers and increase sales. Also, more remaining shelf life at the time of purchase by the consumer can mean a better tasting product, leading to repeat purchases. Finally, as mentioned above, BevCo pays for the cost of expiration occurring at retail stores, which includes the cost of reverse logistics, handling, and disposal. These costs are higher on a per-unit basis at retail stores compared to warehouses mainly due to quantity differences in handling (i.e., individual units or cases versus pallets), as we discuss further in Section 6.4. Hence, by imposing minimum shelf life requirements, BevCo can avoid the higher waste cost at stores for units that would have little chance of selling before expiration.

3 | MODEL

3.1 | Model description

We consider a two-echelon supply chain with a CPG manufacturer selling and distributing a perishable product with a finite deterministic shelf life of N periods from a warehouse

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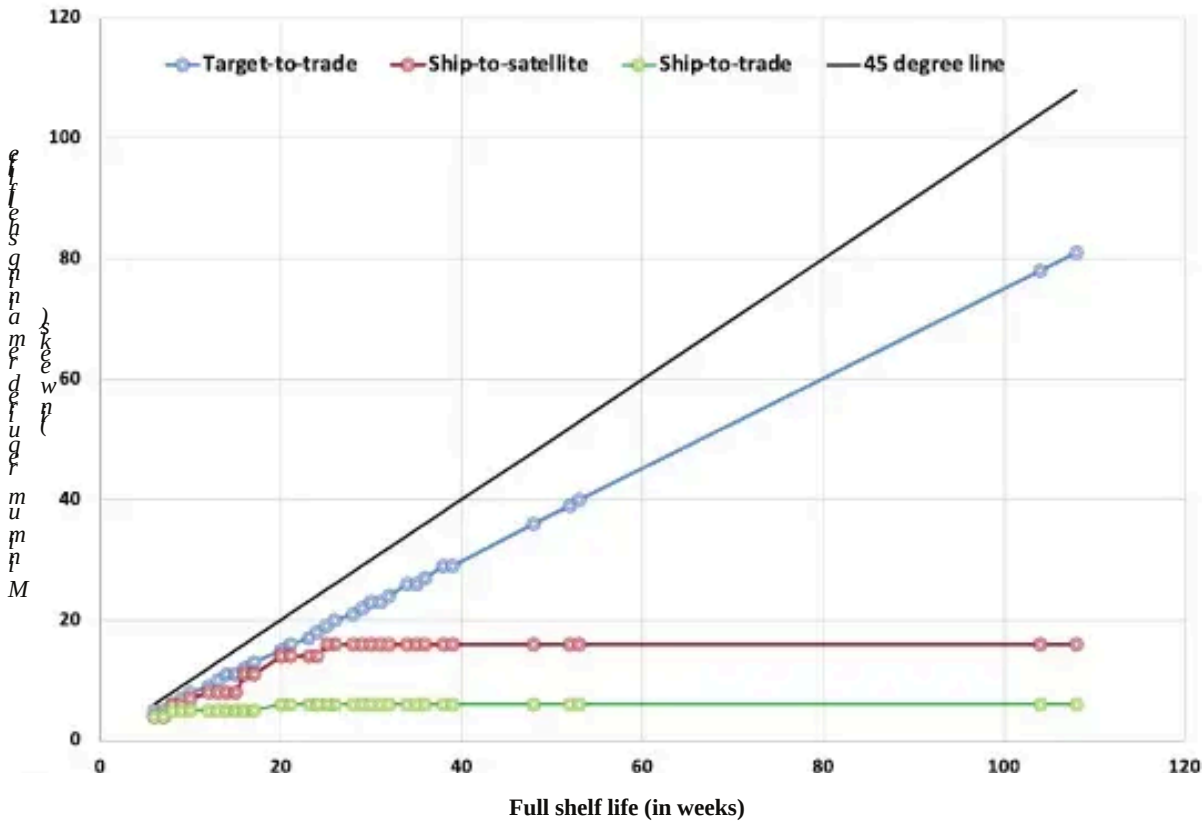


FIGURE 1 Existing shipping rules at BevCo (expressed in remaining shelf life). [Color figure can be viewed at wileyonlinelibrary.com]

to a network of R retail stores. Let $\mathcal{R} = \{1, \ldots, R\}$ denote the set of retail stores. The manufacturer sends shipments to the retail stores periodically, selecting from its available inventory at the warehouse. The manufacturer earns a gross margin of m^r per unit shipped to retail store $r \in \mathcal{R}$. Any product that is unsold after N periods has to be disposed of. If the product expires at the warehouse, the manufacturer incurs a waste cost of c^W. If the product expires at one of the retail stores, the manufacturer gives the retailer a refund and picks up the expired units to dispose of, incurring a waste cost of c^r per expired unit at retailer r, such that $c^r > c^W$ for every $r \in \mathcal{R}$.

We use a discrete-time periodic inventory review model and the timing of events in each period is as follows. At the start of each period, a shipment from the plant is received by the warehouse. Second, the manufacturer observes the remaining inventory at the retail stores and ships units from the warehouse inventory to each retail store. Third, the retail stores receive their shipment from the warehouse (with zero lead time) and the manufacturer picks up the expired units (if any) on the store shelves to dispose of them. Fourth, consumers buy the products from the retail stores. Finally, the cost of waste at the warehouse and at the retail stores is incurred by the manufacturer.

We assume that the product is shipped from the external supplier to the warehouse in pallets of a fixed size p and from the warehouse to the retail stores in cases of a fixed size c, where p is a multiple of c. At the retail store, the product is sold by the unit.

The manufacturer follows a SOF issuing policy with a shipping age threshold $\hat{n}$, that is, the units to ship to retail stores are selected in decreasing order of the inventory's age, excluding units that are older than $\hat{n}$. In other words, units that reach age $\hat{n}+1$ at the warehouse are disposed of, that is, do not get shipped to retail stores. The extreme cases are $\hat{n} = 0$, which means that the warehouse only ships units of age 0 to the store and $\hat{n} = N-1$, which means that all nonexpired units can be shipped to the retail stores.

Let $y_n^{t,r}$ denote the inventory of products of age n at retail store r at the start of period t, and let $q_n^{t,r}$ denote the shipping quantity of age n products from the warehouse to retail store r in period t. Also, let $\mathbf{y}^{r,t} = (0, y_1^{r,t}, \ldots y_{N-1}^{r,t})$ be the inventory vector at retail store r in period t and $\mathbf{q}^{t,r} = (q_0^{r,t}, \ldots q_{N-1}^{r,t})$ denote the shipping quantity vector from the warehouse to retail store r in period t. Note that, by construction, we always have $y_0^{r,t} = 0$ for all t and $r \in \mathcal{R}$. Inventory at the retail stores is replenished using an order-up-to policy; let S^r denote an order-up-to level at retail store $r \in \mathcal{R}$. Given that the product is shipped from the warehouse to the retail stores in cases of size c, the total quantity shipped to retailer store r in period t is $Q^{r,t} = \sum_{n=0}^{N-1} q_n^{r,t} = \frac{(S^r - \sum_{n=1}^{N-1} y_n^{r,t})^+}{c}$. This implies that nothing is shipped in period t if $\sum_{n=1}^{N-1} y_n^{r,t} \geq S^r - \frac{c}{2}$, that is, the retail store inventory must have dropped by at least half-a-case for the retail store to receive a shipment. Note that, in a given period, the inventory at retail store r can be higher than S^r units due to the rounding to the nearest multiple of the case pack size (hence the positive part in the expression).

Let x_n^t be the number of units of age $i \in \{1, \ldots, N-1\}$ in the inventory at the warehouse at the start of period t. The manufacturer also uses an order-up-to policy with level S^W to replenish the warehouse inventory; the units received arrive

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from a (fully reliable) manufacturing plant with a zero-lead time and with their full shelf life. Given that the product is shipped in pallets of size p from the external supplier to the warehouse, the quantity received by the warehouse in period t is $x_0^t = \frac{(S^W - \sum_{n=1}^{\hat{n}} x_n)^+}{p}$. Note that this assumes that units which are older than $\hat{n}$ are ignored from this calculation since they are disposed of. Also, this implies that nothing is received if, in period t, $\sum_{n=1}^{\hat{n}} x_n^t \geq S^W - \frac{p}{2}$, that is, the warehouse inventory must drop below the threshold S^W by more than half-a-pallet in order for a shipment to be received.

The units in the inventory at the warehouse are allocated to the R retail stores, in decreasing order of age up to threshold age $\hat{n}$, following a cyclic allocation method which goes as follows: starting with the units of age $\hat{n}$, one unit is allocated to retail store 1, the next unit to retail store 2, the next one to retail store 3, etc. then possibly back to retail store 1 if $x_{\hat{n}}^t$ is greater than R. Once all the units of age $\hat{n}$ have been allocated, the manufacturer moves on to units of age $\hat{n} - 1$ and the cycle repeats starting with the retail store that was last allocated a unit. This goes on until either all the units in the warehouse have been allocated or all the shipment quantities to retail stores have been met.

The total number of consumers shopping at retail store r in period t, referred to as the *store demand rate* in store r, is a random variable denoted by $\xi^{r,t}$, which is independently and identically distributed across periods with mean λ^r for $r \in \mathcal{R}$. We assume that a proportion $\alpha \in [0,1]$ of these consumers depletes the inventory from the store shelves in a FIFO fashion, with the remaining $1 - \alpha$ depleting it in a last-in-first-out (LIFO) fashion (a similar assumption is made in Minner & Transchel, 2017, and Akkaş & Honhon, 2022). The two extreme cases are $\alpha = 1$, which corresponds to pure FIFO inventory depletion, and $\alpha = 0$, which corresponds to pure LIFO inventory depletion. Let $D_n^{r,t}(\mathbf{z}^{r,t}, \xi^{r,t}; \alpha)$ denote the store demand for units of age n in period t at retail store r as a function of the postdelivery vector of inventory levels at retail store r in period t, denoted by $\mathbf{z}^{r,t} = \mathbf{y}^{r,t} + \mathbf{q}^{r,t}$. We have $D_n^{r,t}(\mathbf{z}^{r,t}, \xi^{r,t}; \alpha) = \left(\alpha\xi^{r,t} - \sum_{j=n+1}^{N-1} z_j^{r,t}\right)^+ + \left((\xi^{r,t} - \alpha\xi^{r,t}) - \sum_{j=0}^{n-1} z_j^{r,t}\right)^+$ for $n = 0, \ldots, N-1$. Note that the rounding is necessary because we are working with discrete quantities.

The inventory balance equations are as follows:

$$x_n^{t+1} = \begin{cases} x_0^t - \sum_{r=1}^{R} q_0^{r,t} & n = 1 \\ x_{n-1}^t - \sum_{r=1}^{R} q_{n-1}^{r,t} & n = 2, \ldots, \hat{n} \\ 0 & n = \hat{n}+1, \ldots, N-1 \end{cases},$$

$$y_n^{r,t+1} = \begin{cases} y_0^{r,t} - D_0^{r,t}(\mathbf{q}^{r,t} + \mathbf{y}^{r,t}, \xi^{r,t}; \alpha)^+ & n = 1 \\ y_{n-1}^{r,t} + q_{n-1}^{r,t} - D_{n-1}^{r,t}(\mathbf{q}^{r,t} + \mathbf{y}^{r,t}, \xi^{r,t}; \alpha)^+ & n = 2, \ldots, N-1 \end{cases}, r \in \mathcal{R}$$

ALGORITHM 1 Shipment quantities

For $n = 0, \ldots, N-1$ and $r \in \mathcal{R}$, $q_n^{r,t} = 0$

For $r \in \mathcal{R}$, $Q^r = S^r - \sum_{n=1}^{N-1} y_n^{r,t}$

$r = 1$.

for $n = \hat{n}, \hat{n}-1, \ldots, 0$ **do**

$X_n = x_n^t$,

while $X_n > 0$ and $\sum_{r=1}^{R} Q^r > 0$ **do**

if $Q^r > 0$ **then**

$q_n^{r,t} := q_n^{r,t} + 1$; $Q^r := Q^r - 1$; $X_n := X_n - 1$.

end if

$r := \mod(r, R) + 1$

end while

end for

The profit function for the manufacturer in period t is as follows:

$$\Pi(\hat{n}, \mathbf{x}^t, \mathbf{y}^{1,t}, \ldots \mathbf{y}^{R,t}, \xi^t; \alpha) = \sum_{r=1}^{R} m^r \sum_{n=0}^{N-1} q_n^{r,t} - c^W \left(x_{\hat{n}}^t - \sum_{r=1}^{R} q_{\hat{n}}^{r,t}\right) - \sum_{r=1}^{R} c^r \left(y_{N-1}^{r,t} + q_{N-1}^{r,t} - D_{N-1}^{r,t}(\mathbf{q}^{r,t} + \mathbf{y}^{r,t}, \xi^{r,t}; \alpha)^+\right) \quad (1)$$

where $\mathbf{q}^{r,t}$ is given by Algorithm 1. In (1), the first term is the gross margin from the units shipped to the retail store, the second term is the cost for the units expired at the warehouse, and the third term is the cost for the units expired at the store.

The manufacturer's problem is to determine the threshold age $\hat{n}$ beyond which units of inventory at the warehouse are not shipped to the retail store in order to maximize the long-run average profit over an infinite time horizon denoted by $\bar{\Pi}(\hat{n})$. Hence, the problem is to find the shipment policy π^* such that

$$\pi^* = \max_{\hat{n} \in \{0, \ldots, N-1\}} \bar{\Pi}(\hat{n}) \quad \text{where}$$

$$\bar{\Pi}(\hat{n}) = \lim_{T \to \infty} \frac{\sum_{t=1}^{T} \Pi(\hat{n}, \mathbf{x}^t, \mathbf{y}^{1,t}, \ldots \mathbf{y}^{R,t}, \xi^t; \alpha)}{T}. \quad (2)$$

We provide details on how to solve this problem in the Supporting Information Appendix B.

3.2 | Examples

In this section, we demonstrate the trade-offs involved in the manufacturer's problem through two numerical examples, each representing a different type of profit function.

Example 1. Let $N = 13$, $R = 5$, $p = 30$, $c = 6$, $m^r = 0.5$, $c^W = 0.8$, and $c^r = 10.5$ for $r \in \mathcal{R}$. Consumer demand has a Poisson distribution with mean $(\lambda^1, \lambda^2, \lambda^3, \lambda^4, \lambda^5) =$

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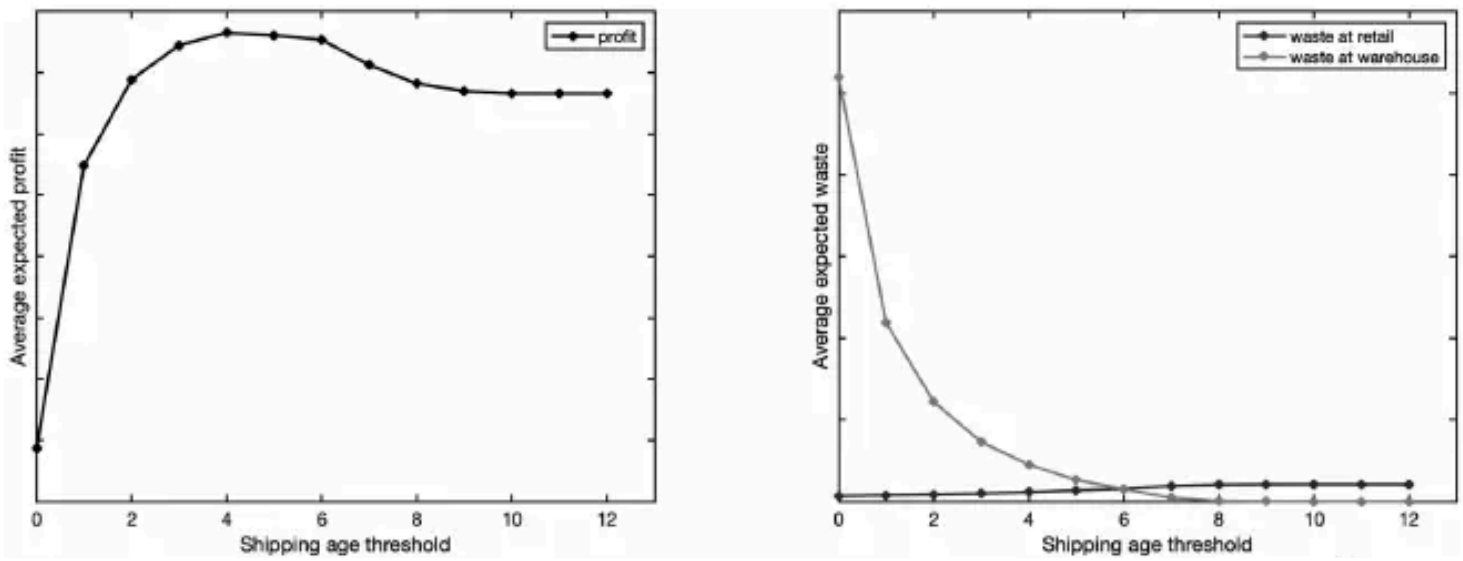


FIGURE 2 Profit and waste as a function of the shipping age threshold in Example 1.

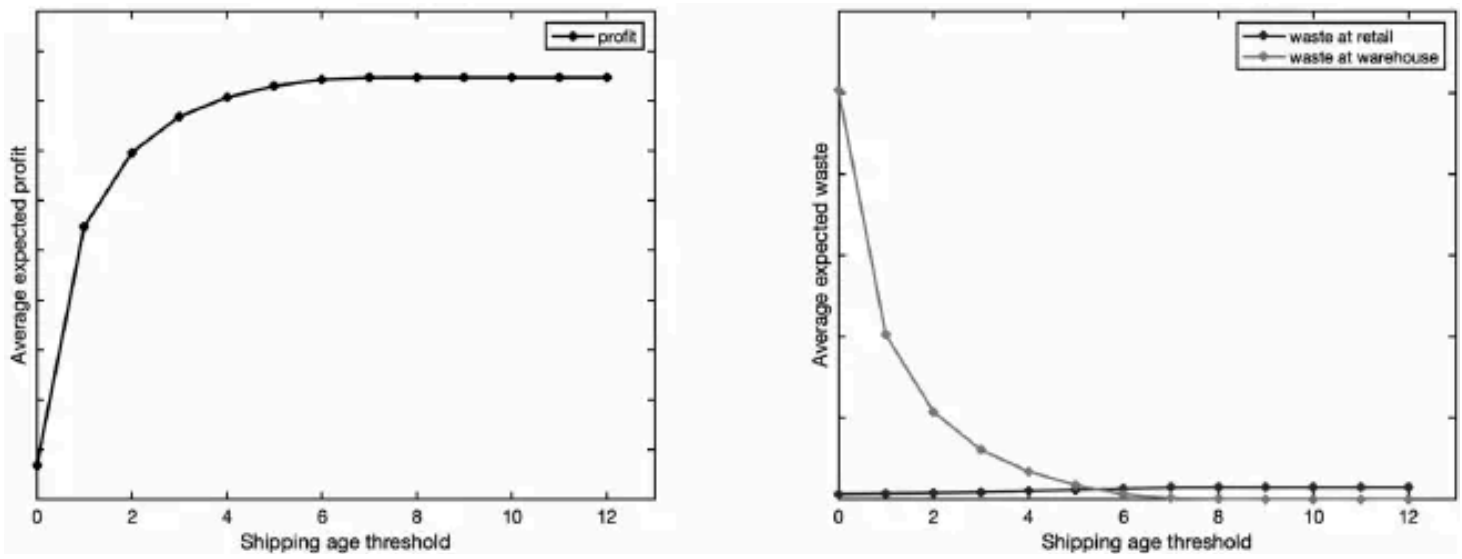


FIGURE 3 Profit and waste as a function of the shipping age threshold in Example 2.

(0.2, 0.3, 0.8, 2.3, 3.4) and inventory depletion at all five stores is an equal mix of FIFO and LIFO, that is, □ =0.5.

Figure 2 (left) represents profit as a function of the shipping age threshold. We see a clear peak at a value of $\hat{n} = 4 < 12 = N - 1$; thus, any inventory that is older than four periods should not be shipped to retail stores. In Figure 2 (right), we see that, as the shipping age threshold increases, warehouse waste decreases and the decrease is occurring at a slower and slower pace (i.e., the function is convex decreasing). Similarly, retail waste increases and the increase is occurring at a slower and slower pace (i.e., the function is concave increasing). Thus, the manufacturer faces a trade-off between the two types of waste.

Example 2. This example uses the same parameters as in Example 1 except $c^r = 2$ for r □□ and $(□^1, □^2, □^3, □^4, □^5) = (0.3, 0.4, 1, 3.0, 4.5)$. Figure 3 (left) represents profit as a function of the shipping age threshold. We see that profit becomes flat past a value of the shipping age threshold $\hat{n} = 9$ and until a maximum of $\hat{n} = 12 = N - 1$. This means that a pure SOF policy is optimal, that is, there is no need to impose a shipping age threshold. Compared to Example 1, the manufacturer faces the same trade-off between the warehouse and retail-level waste; however, because retail waste is less costly (c^r is lower than c^W) and less likely to occur (because □ for r □□ are higher) in this example, the manufacturer can afford to send older units to the retail stores; as a result, the optimal shipping age is higher. In fact, there is a very low probability of a unit aging past nine periods, which is why the manufacturer can follow a pure SOF issuing policy.

3.3 | Analytical results

We present our results for the special case of $p = c = 1$ to avoid any discreetness effect and for ease of exposition. We also assume that inventory at the retail store is depleted in a FIFO fashion (i.e., □ =1). All proofs are presented in the Supporting Information Appendix A.

Let $□^{W,t} = x^t_{\hat{n}} - \sum_{r=1}^{R} q^{r,t}_{\hat{n}}$ be the waste quantity at the manufacturer's warehouse in period t, and let $□^{r,t} = [y^{r,t}_{N-1} + q^{r,t}_{N-1} - D^{r,t}_{N-1}(\mathbf{q}^{r,t} + \mathbf{y}^{r,t}_{N-2}, □^t, □)]^+$ be the waste quantity at

retail store $r \in \mathcal{R}$ in period t. Let $\bar{\Delta}^W(\hat{n}) = \lim_{T\to\infty} \sum_{t=1}^{T} \Delta^{W,t}$ and $\bar{\Delta}^r(\hat{n}) = \lim_{T\to\infty} \sum_{t=1}^{T} \Delta^{r,t}$ be the long-run average waste quantities at the manufacturer's warehouse and at retail store $r \in \mathcal{R}$, respectively, when the shipping age threshold is $\hat{n}$. The profit function in period t can be written as follows:

$$\Pi(\mathbf{x}^t, \mathbf{y}^{1,t}, \ldots \mathbf{y}^{R,t}, \mathbf{q}; \mathcal{I}, \hat{n}) = \sum_{r=1}^{R} m^r Q^{r,t} - c^W_{\hat{n}} \Delta^{W,t} - \sum_{r=1}^{R} c^r \Delta^{r,t}.$$

We show that, as the maximum shipping age threshold $\hat{n}$ increases, waste at the retail stores increases (weakly) but waste at the manufacturer's warehouse and overall waste (weakly) decreases.

Proposition 1. *$\bar{\Delta}^W(\hat{n})$ is nonincreasing in $\hat{n}$. For all $r \in \mathcal{R}$, $\bar{\Delta}^r(\hat{n})$ is nondecreasing in $\hat{n}$. $\bar{\Delta}^r(\hat{n}) + \bar{\Delta}^W(\hat{n})$ is nonincreasing in $\hat{n}$.*

The intuition behind Proposition 1 is that, when the age-threshold $\hat{n}$ is increased, the warehouse has more periods to sell the inventory before it reaches the end of its shelf life; therefore, waste at the warehouse is less. However, the units reach the retail stores at a more advanced age (on average) so they have fewer opportunities to sell to the end consumers before expiring, leading to more waste at the retail stores. Yet, a lower threshold results in more waste overall because the manufacturer gives up on selling some of the units sooner, by not giving them a chance to reach the store shelves.

In our problem, because the cost of waste is higher at the retail stores than at the warehouse, it may make sense for the profit-maximizing manufacturer to set an age-threshold that is strictly less than the product's shelf life. With our next result, we provide a sufficient condition for the optimal shipping age threshold to be $N-1$, which amounts to the pure SOF issuing policy (i.e., no threshold) being optimal.

Proposition 2. *If $c^W \geq c^r - m^r$ for all $r \in \mathcal{R}$, then the optimal shipping age threshold is $\hat{n}^* = N-1$.*

If the condition of Proposition 2 is not satisfied, that is, if $c^W < c^r - m^r$ for some $r \in \mathcal{R}$, then it is possible that the optimal shipping age threshold is strictly less than $N-1$, which means that it makes sense for the profit-maximizing manufacturer to increase total waste in order to increase profits. It is noteworthy that this condition does not involve the shelf life N or any information about the demand distribution, only the cost parameters.

With our final result, we characterize the profit function and show that when warehouse waste goes down at a decreasing rate and retail waste (at all stores) goes up at a decreasing rate as a function of the shipping age threshold, the profit function has a single peak. Let $\Delta\bar{\Delta}^W(\hat{n}) = \bar{\Delta}^W(\hat{n}) - \bar{\Delta}^W(\hat{n}-1)$ and $\Delta\bar{\Delta}^r(\hat{n}) = \bar{\Delta}^r(\hat{n}) - \bar{\Delta}^r(\hat{n}-1)$ for $r \in \mathcal{R}$ denote the first differences in long-run waste quantities at the warehouse and retail store, respectively.

Proposition 3. *If (i) $\Delta\bar{\Delta}^W(\hat{n}+1) \geq \Delta\bar{\Delta}^W(\hat{n})$ and (ii) $\Delta\bar{\Delta}^r(\hat{n}+1) \geq \Delta\bar{\Delta}^r(\hat{n})$ for all $r \in \mathcal{R}$, then $\bar{\Pi}(\hat{n})$ is unimodal in $\hat{n}$.*

The result in Proposition 3 is stated in terms of first differences due to the discrete nature of the shipping age threshold $\hat{n}$, but the equivalent continuous condition is that the long-run average of waste quantity at the warehouse is decreasing convex in $\hat{n}$ and the long-run average of the waste quantity at the retail store is increasing concave in $\hat{n}$. If the conditions of Proposition 3 are satisfied—and we find numerically that this is the case (see Examples 1 and 2)—there exists a unique optimal value $\hat{n}^*$ of the shipping age threshold.

4 | NUMERICAL STUDY

4.1 | Overview of our dataset

The dataset from our industry collaborator consists of annual sales and returns, shelf life, case size, cost-of-goods-sold (COGS), and wholesale price data on 455 SKUs.

BevCo also provided us with an estimate of the sales and delivery cost as well as the disposal cost at retail stores (the disposal cost is primarily driven by the handling cost; see a discussion in Section 5.4). We computed the gross margin for each SKU as the wholesale price minus the COGS, minus the sales and delivery cost. We computed the retail waste cost as the wholesale price plus the retail-level disposal cost and made the warehouse waste cost equal to the COGS (we considered the warehouse handling cost to be negligible, relative to that at the stores, due to the use of forklifts and pallet-level handling). Figure 4 shows a sample of our dataset including the variables we compute. Table 1 presents the summary statistics of our data across all SKUs. Note that the numbers for the store demand rate in the table are the percentiles across different SKUs for the median store demand across stores.

4.2 | Implementation of our method

We utilize BevCo's dataset for two purposes. First, we conduct a case study (Section 4.3), that is, we use our method to calculate the optimal shipping age threshold for all the SKUs from the dataset. Second, we run a sensitivity analysis (Supporting Information Appendix C) where we study the impact on the optimal shipping age threshold of all the model parameters. In this subsection, we describe the assumptions we make for both tasks.

While our dataset covers over 60,000 stores and 200 warehouses, in our case study and sensitivity analysis, we assume that there are five retail stores ($R = 5$) supplied by a single

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SKU ID	shelf life (in weeks)	case size (in units)	demand (units/week) store 1 (10th percentile)	store 2 (25th percentile)	store 3 (50th percentile)	store 4 (75th percentile)	store 5 (90th percentile)	gross margin	waste cost at warehouse	waste cost at retail stores	existing threshold (in weeks) ship-to-trade	existing threshold (in weeks) maximum shipping age
1	13	12	1.38	2.98	6.00	12.22	21.69	$ 0.76	$ 0.43	$ 1.62	6	7
2	39	4	0.77	3.27	9.00	20.38	40.15	$ 0.62	$ 1.18	$ 3.09	6	33
3	13	15	1.60	2.77	4.84	8.65	17.31	$ 0.71	$ 0.43	$ 1.48	6	7
4	26	4	0.54	1.85	6.77	15.54	29.57	$ 0.51	$ 1.25	$ 3.05	6	20
5	16	8	0.92	2.44	6.31	16.46	40.91	$ 0.26	$ 0.83	$ 1.74	6	10
6	14	8	0.40	1.09	3.23	10.46	26.46	$ 0.32	$ 0.82	$ 1.79	6	8
7	26	4	0.08	0.15	0.85	1.08	1.86	$ 0.50	$ 1.23	$ 3.02	6	20
8	12	8	0.31	0.92	2.98	9.48	18.31	$ 0.35	$ 0.77	$ 1.77	6	6
9	12	8	0.31	0.92	3.23	8.31	18.00	$ 0.31	$ 0.78	$ 1.73	6	6
10	12	8	0.46	1.38	4.00	12.92	29.63	$ 0.35	$ 0.79	$ 1.79	6	6

FIGURE 4 Sample of the dataset. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 1 Summary statistics of the parameters in our dataset.

Percentile	5%	10%	50%	90%	95%
Shelf life, N (in weeks)	13	13	20	52	78
Case size, c (in units per case)	1	2	8	24	24
Store demand rate, $\square^r$ (in units per week)	0.49	0.65	1.85	5.54	8.25
Gross margin, m^r (in $/unit)	0.147	0.224	0.580	1.335	1.613
Waste cost at warehouse, c^W (in $/unit)	0.252	0.272	0.823	2.865	4.715
Waste cost at retail, c^r (in $/unit)	0.931	0.981	1.778	6.676	10.629

warehouse in order to keep computational times low. To set the values of the store demand rates of each SKU at the five stores ($\square^1, \square^2, \square^3, \square^4, \square^5$), we used the 10th, 25th, 50th, 75th, and 90th percentiles of sales to stores, net of returns, across all the stores in the dataset for each SKU. Also, we assume identical gross margin and waste cost across all five stores, as we received these data at the SKU level.

We set the order-up-to level at the retail stores as $S^r = \square^r + 2.326\sqrt{\square^r}$, which is the formula for the order-up-to level from periodic review inventory management assuming a 99% service level and Poisson demand with the value rounded up to a multiple of the case pack size. We set $S^W = \left\lceil \frac{\sum_{r=1}^{5} S^r}{p} \right\rceil p$ that is, the order-up-to level at the warehouse is equal to the sum of the order-up-to levels at the all the retail stores, rounded up to the next multiple of the pallet size. This is to guarantee that the warehouse always has sufficient inventory to send full shipments to all the retail stores. Further, we assume that the pallet size is five times the case size as we consider five stores in our analysis; this ensures that we observe replenishment increments of a reasonable magnitude. Also, unless otherwise noted, we assume a 50–50% mix of the FIFO and LIFO inventory depletion by consumers at the retail stores (i.e., $\square = 0.5$).

All figures in Section 4.3.2 and Supporting Information Appendix C show two different values for the optimal shipping age threshold, specifically the *minimum* and *maximum*, defined as the lowest and highest values of the shipping age threshold that achieve at least 99.95% of the highest expected profit value. This flexibility in interpreting the concept of "optimal shipping age threshold" is necessary because we use simulation to estimate profit due to the prohibitive size of the state space for most problem instances. Profit and waste values are computed based on the median optimal threshold, that is, the value that is halfway between the minimum and maximum threshold values.

4.3 | Case study

In this section, we apply our methodology to the dataset from our industry collaborator described in Section 4.1. Specifically, we compute the optimal shipping age threshold for each of BevCo's 455 SKUs, separately, under the assumption of FIFO and LIFO inventory depletion at retail stores.

We find that for 90% of BevCo's SKUs, the current practice threshold falls between the minimum and maximum (profit-maximizing) shipping age thresholds under FIFO. We obtain similar results under the LIFO assumption: for 91% of the SKUs, the current threshold is within the optimum range of the profit-maximizing thresholds. Thus, we conclude that BevCo's existing thresholds are appropriate for most of their SKUs.

Next, we focus our analysis on the 47 (39) SKUs where BevCo's current practice thresholds are suboptimal under FIFO (LIFO) and first examine the characteristics of these SKUs. We find that for 39 (23) out of these, BevCo's current practice threshold is lower than the optimal range, under FIFO (LIFO), which means that BevCo should send older units to the retail stores. In Table 2, we present summary statistics for the SKUs in our dataset, which we separate into three groups depending on whether BevCo's current practice threshold falls: within the optimal range (Group 1), below the optimum range (Group 2), or above the optimal range (Group 3). We see that the SKUs for which BevCo's current practice is suboptimal are typically slow-moving items (i.e., low $\square^\beta$). The ones for which BevCo's current practice is too conservative (Group 2) typically also have shorter shelf lives, especially relative to the expected store demand rate. Group 2 SKUs are also characterized by a higher ratio $\frac{c^r}{c^W}$ and a lower ratio $\frac{c^r}{m^r}$. Based on Proposition 2, we would expect the two ratios to have opposite effects on the optimal shipping thresholds; which suggests that the latter ratio has a dominating effect.

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TABLE 2 SKU summary statistics based on current practice threshold versus optimal range.

Group	Number of SKUs	N	σ^3	m^r	c^W	c^r	c^r/c^W	c^r/m^r	N/σ^3
1	408	28.8	3.0	0.8	1.6	3.7	2.6	7.0	23.1
Within range	(416)	(28.4)	(3.0)	(0.8)	(1.6)	(3.7)	(2.6)	(6.9)	(22.8)
2	39	13.7	1.5	0.5	0.4	1.4	3.5	2.8	13.4
Below range	(23)	(13.7)	(1.4)	(0.5)	(0.3)	(1.1)	(3.5)	(2.4)	(13.5)
3	8	24.3	1.1	0.3	0.5	1.1	2.5	5.6	25.0
Above range	(16)	(21.7)	(1.3)	(0.4)	(0.7)	(1.7)	(2.6)	(5.0)	(21.2)
All	455	27.5	2.8	0.7	1.5	3.5	2.7	6.6	22.3

Note: Numbers represent the averages for `FIFO` (`LIFO`).
Abbreviations: FIFO, first-in-first-out policy; LIFO, last-in-first-out policy.

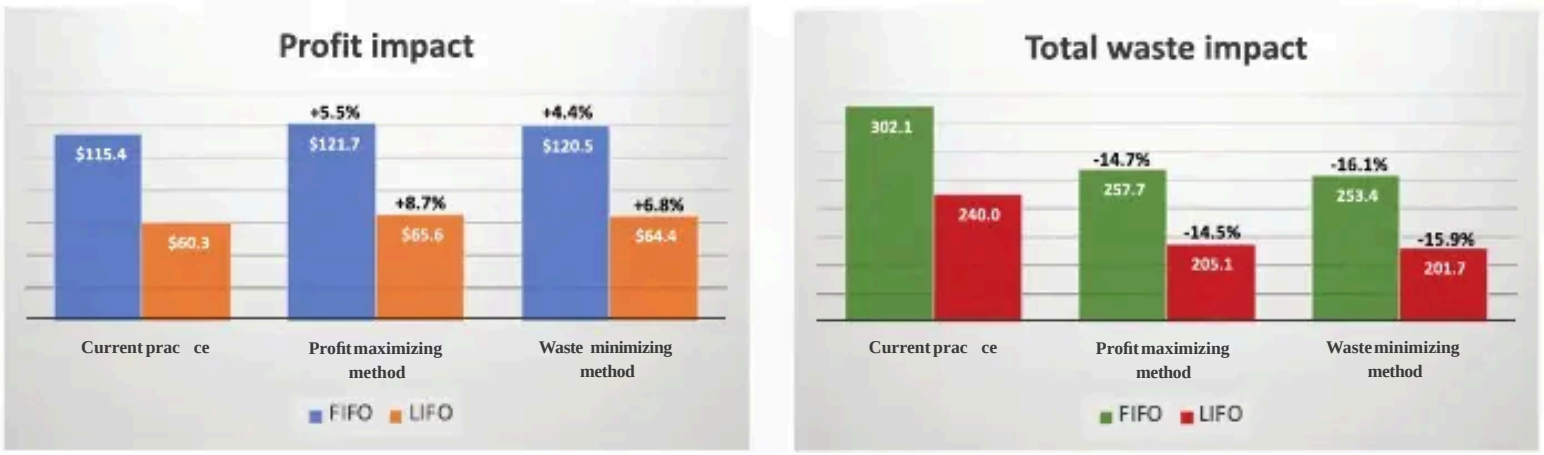


FIGURE 5 Performance comparison of three methods for SKUs with suboptimal thresholds (profit and waste numbers represent weekly figures for the five stores considered). [Color figure can be viewed at wileyonlinelibrary.com]

As described in Section 3, the optimal shipping age thresholds we calculate are obtained from a profit-maximization approach. We now consider two alternative approaches as benchmarks: the current practice at BevCo and a waste-minimizing approach. As described in Section 2, BevCo currently sets shipping age thresholds for most SKUs as their shelf life minus 6 weeks. With the waste-minimizing approach, we minimize the total waste at the warehouse and retail stores combined, which by Proposition 1, is achieved by using a pure SOF policy without any thresholds.

Under all three methods (i.e., profit maximizing, current practice, and waste minimizing), we compute the weekly expected profit and the weekly expected total waste for each SKU and show these in Figure 5. We see that, compared to current practice, the profit-maximizing approach reduces waste by a 14.7% (14.5%), which is comparable to the waste reduction of 16.1% (15.9%) from the waste-minimizing approach, under FIFO (LIFO) depletion at retail stores. By definition, the profit-maximizing method outperforms the waste-minimizing method in terms of expected profit, resulting in a 5.5% (8.7%) profit increase compared to current practice versus 4.4% (6.8%) for the waste-minimizing method under FIFO (LIFO).

In order to get a sense of the magnitude of potential profit improvement on a yearly basis for BevCo, we extrapolate the numbers above by multiplying the difference in profits between the profit-maximizing method and BevCo's current practice by the annual actual sales for each SKU across all stores and divide this result by the total order quantity across the five stores in our analysis. Summing these numbers across all SKUs with a suboptimal current practice threshold, we obtain annual profit differences of $292,561 and $125,269, respectively, under FIFO and LIFO. Note that these figures are only estimates of how much BevCo could save by using our method compared to a situation of full compliance with the current practice rules. Actual potential savings could be different depending on actual compliance.

We follow the same extrapolation approach to assess the impact on total waste. For this, we use product weight data which we extract from product descriptions (provided in ounces or liters). We find that our profit maximization method, compared to BevCo's current practice, offers 1846 and 559 truckloads worth of annual waste reduction under the FIFO and LIFO, respectively, assuming a 20-ton load per truck.[3] This method offers substantial waste reduction because it generally finds that the shipping age threshold can be older than what BevCo's current practice dictates.

4.3.1 | Optimizing the common threshold

In practice, manufacturers usually set a common threshold across all of their SKUs, which is typically expressed as a fixed number of weeks remaining until expiration, as is the

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FIGURE 6 Profit impact of deviating from the optimal common threshold. [Color figure can be viewed at wileyonlinelibrary.com]

case with BevCo and their 6-week *ship-to-trade* rules. We believe that, for most manufacturers, this is primarily due to a lack of sophistication and methodology to determine SKU-specific shipping rules, but there are also some operational simplicity benefits of using a one-size-fits-all approach, such as for training warehouse employees and the salesforce about shelf life management expectations.

In this section, we consider the case of a common threshold across all SKUs and find the one that maximizes total (extrapolated) profits across all 455 SKUs in our database.[4] Assuming an equal mix of FIFO and LIFO at retail stores (i.e., $\square$ =0.5), we find that the profit-maximizing common threshold is equal to 6 weeks of remaining shelf life, which is BevCo's current practice of the *ship-to-trade* rule. Reducing the common threshold to 5 weeks would result in a profit loss of $168,447 and increasing it to 7 weeks would result in a loss of $781,465 (see Figure 6). In other words, if BevCo is to keep a threshold common to all of its SKUs, it should not change its current practice of a 6-week *ship-to-trade* rule. However, as our analysis from the previous section demonstrates, the company can significantly increase profits and reduce waste by implementing SKU-specific shipping age thresholds (Figure 5).

4.3.2 | Comparison to current rules

Here, we compare the optimal shipping age threshold values to the current practice at BevCo as described in Section 2. In Figure 7, we do so in four scenarios (corresponding to different values of the expected store demand rates and retail waste cost, chosen from the values in Table 1). We see that, when the shelf life of the product is long enough, BevCo's current practice falls into the optimal range of shipping age threshold; however, for shorter shelf life products, it falls outside of the range. Specifically, BevCo's policy appears too conservative for products with a high retail waste cost (Figure 7c,d), that is, too many products are wasted at the warehouse; on the other hand, for products with a low retail waste cost (Figure 7a,b), BevCo's policy is too loose, that is, too many old products are shipped to retail stores.

5 | BROADER IMPLICATIONS FOR PRACTICE

In this section, we discuss various aspects of our problem in order to guide practitioners in implementing our methodology.

5.1 | Freshness and quality

Expiration dates on CPG are usually treated as a hard constraint by both researchers and practitioners. In the perishable inventory management literature, products with fixed shelf lives typically only incur an expiration cost past their expiration date. In practice, retailers generally pull products off their shelves once they expire, even usually slightly prior to reaching their label date. Similarly, many consumers dispose of products they purchased when they are past their expiration date, treating the label date as a binary indicator of the product's safety to consume. In fact, in the United States, with the exception of infant formula, date labels are not regulated and are chosen by manufacturers to help consumers and retailers decide when food is of the *best quality*, which in itself is a subjective concept (USDA, 2022). For example, for canned products, expiration dates are usually set very conservatively (e.g., about 1 year when in fact, most products retain their taste attributes for several years) for fear that consumers would perceive long shelf lives as an indication that the product is highly processed with strong preservatives. On the other hand, some fresh products with relatively shorter shelf lives (e.g., pastries, granola bars) could experience a taste deterioration before their expiration date. As a result, consumers who taste the product near expiration may choose not to purchase the product again in the future. Another scenario that can negatively impact repeat purchases is products

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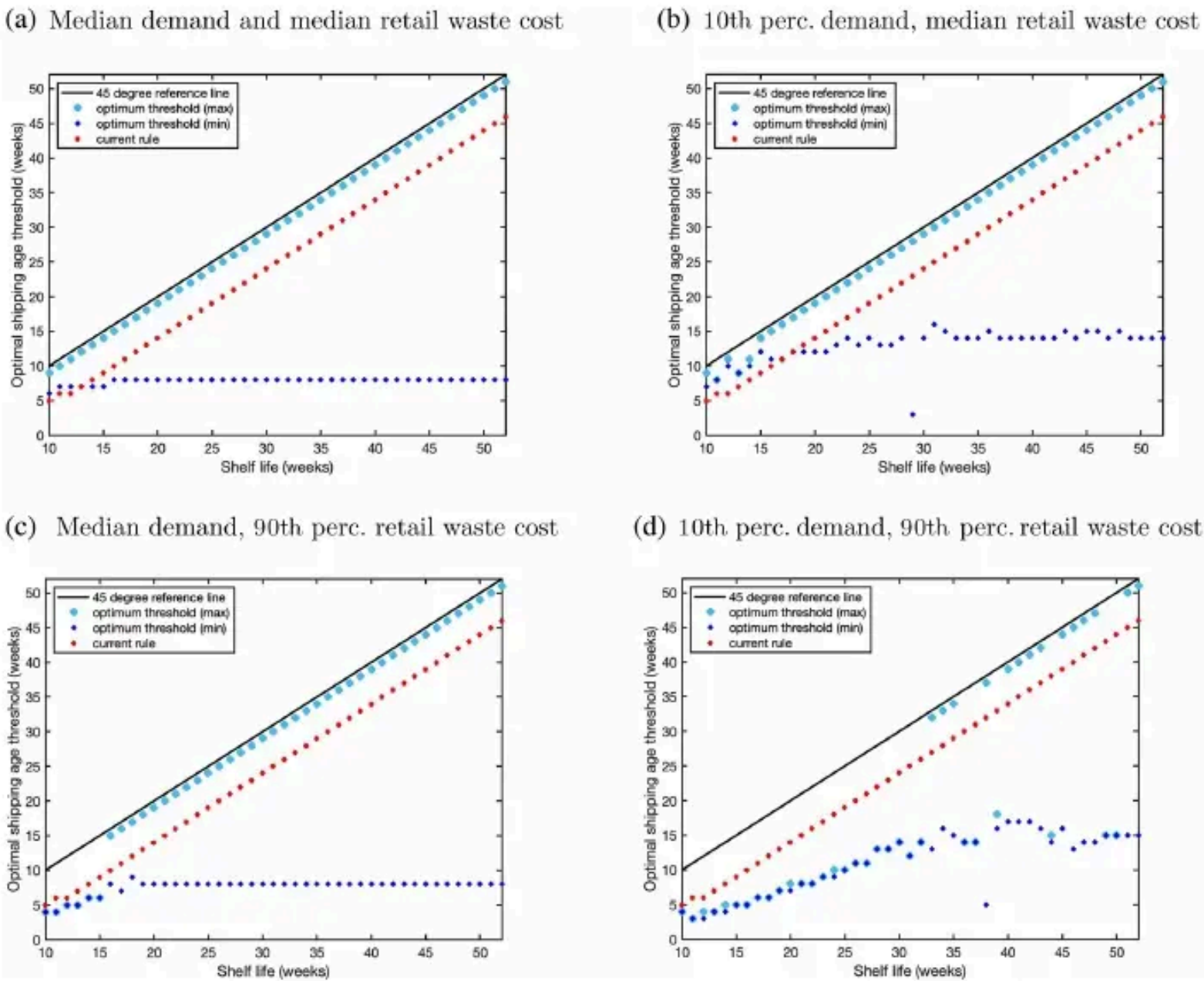


FIGURE 7 Impact of shelf life on the optimum shipping age threshold and the current practice at BevCo. [Color figure can be viewed at wileyonlinelibrary.com]

expiring in consumers' pantry or fridge because they were purchased with too little remaining shelf life. This can reflect poorly on the retailer and the manufacturer, making the consumer switch to another store or another brand. In our model, we have chosen not to include a *staleness cost* to capture the effect that aging inventory could have on consumer demand. This is because estimating such a cost would require a comprehensive study of its own. Further, the staleness effect on consumer perception of quality is likely not strong for BevCo products, which are mostly carbonated soft drinks and because their *best taste limits* are set conservatively with a median shelf life of 20 weeks.

5.2 | Assumptions

Our analysis assumes that the warehouse follows a SOF inventory issuing policy. While this policy is in place at BevCo, in practice, it is not always complied with. In an audit study conducted at one particular warehouse, BevCo found that approximately half of the SKUs did not show signs of having been appropriately *rotated* , that is, sequenced in a way that makes it easier to pick the oldest units first. In theory, it would be possible to optimize maximum shipping age thresholds under other inventory issuing policies (e.g., Ship-Freshest-First or Random Pick); however, our recommendation to warehouse managers is to enforce the SOF policy because accommodating suboptimal behavior does not send the right signal to managers who should instead discourage this noncompliant behavior.

We make the simplifying assumption that the products reach the warehouse from the plant with their full shelf life remaining. Although, in practice, this is not always the case, at BevCo, most products arrive at satellite warehouses essentially with their full shelf lives. BevCo has a *deep storage* area at its plant warehouses that typically store excess production, which is utilized to pool inventory for slow-moving items or to store products when transportation and production plans are unsynchronized due to delays in production. To avoid double handling cost, BevCo has a policy to move most products directly from the production line to trucks. To facilitate this, the last step of production, namely palletizing,

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is designed to end at the loading docks. Plant managers are evaluated on a *direct shipment* metric, which measures the total percentage of products directly shipped to satellite warehouses from production lines, with most plants achieving around 80% on this metric. Thus, for most products, our assumption is a reasonable one. Our conversations with another CPG manufacturer (selling primarily snacks) reveal that double handing is a major concern for them as well; thus, they also try to *direct ship* as much as they can. This concern is most likely prevalent across the CPG industry, which is known to operate with low margins, making operational efficiencies crucial for financial performance.

Our analysis also assumes that retail stores follow an order-up-to inventory policy, which is a reasonable assumption because inventory is often replenished up to the product's shelf capacity at retail stores (Akkaş, 2019). At BevCo, this is often referred to as the *fill the hole* policy. Because shelf capacity data at retail stores are not available (neither in our dataset nor for BevCo), we use the target inventory levels from the periodic review inventory management literature for non-perishable products (Zipkin, 2000), as, to this date, there does not exist closed-form solutions to the equivalent problem for perishable inventory (Karaesmen et al., 2011). Note that inventory replenishment only occurs when the inventory level drops below half a case, as we explained in Section 4. This is consistent with the shelf space allocation policy called the *pack-and-a-half-rule*, which requires that the shelf space allocated for a product should be at least 50% larger than its case size (Eroglu et al., 2018). Treating half-a-case as a reorder point allows us to accommodate real-world scenarios where not every product is replenished at the store's scheduled visit (i.e., if the shelf is still sufficiently full, an order is typically not placed for a given product); similarly, not every store serviced by a warehouse follows the same ordering frequency.

BevCo's actual inventory replenishment policy at the warehouse level is complex to replicate. Products are to be ordered in increments of pallets, layers, or cases, depending on the SKU's sales velocity. There is no fixed increment as replenishment quantities are based on a forecast measured in days-of-supply, which can be inflated to obtain full truckloads. Thus, as we do for the retail replenishment policy, we make the reasonable assumption of an order-up-to replenishment policy from the manufacturing plant to the warehouse. We explored whether incorporating the order-up-to level at the warehouse as a decision variable in our analysis adds any value to our analysis and found, from numerical testing, that joint optimization of the shipping age threshold and the order-up-to level does not change the optimal threshold. Therefore, we consider the order-up-to level at the warehouse as an exogenous parameter in our analysis. We compute the order-up-to level as the sum of the order up-to levels across all retail stores, expressed in pallet increments. We set the pallet size equal to five times the case size, which is also the number of stores included in our multiretailer analysis. Using actual pallet sizes is not appropriate, because (i) a warehouse in reality can serve 100 + stores; therefore, actual pallet sizes would represent too much inventory for five stores, leading to unrealistically high waste numbers; and (ii) the warehouse does not necessarily receive products in pallets, as explained earlier.

Finally, we assume that the warehouse inventory is distributed to the retailers in a cyclic manner on a per-unit basis, as explained in Section 3.1. We argue that this is a reasonable, equitable, and common assumption (Maass & Daskin, 2017; Raa & Aghezzaf, 2009) that is consistent with BevCo's practice as they currently do not allocate inventory strategically (e.g., based on sales volume, importance of its customers, etc.). Other possible allocation rules include random distribution of inventory (i.e., each unit of inventory is randomly assigned to a store) or a volume-based allocation method (i.e., high-volume stores with better selling capabilities get the oldest products while low-volume stores get the newest).

5.3 | Implementation and execution

Applying our method to BevCo's 455 SKUs, we find that only 16 (26) SKUs have unique thresholds under FIFO (LIFO); for the remaining SKUs, there exists a range of optimal shipping age threshold values that provide (close to) maximum expected profit. Because practical execution of the shipping strategy requires a single threshold, we propose to select the median value inside the optimal range. Further, for 27 (38) of the SKUs, we find that the maximum value of the optimal threshold is $N - 1$ under FIFO (LIFO). For these SKUs, there is no significant profit benefit of establishing a maximum shipping age threshold, that is, a pure SOF issuing policy is optimal.

Our method assumes an order-up-to level for replenishment at both the warehouse and the retail stores. If another method is used in practice, our method can be modified to adapt to the reality of the replenishment system. We also assume that the manufacturer has knowledge of the demand distribution at all retail stores. In practice, this distribution can be inferred from past sales data or shipment to stores data.

Our analysis has outlined the benefits of SKU-specific shipping age thresholds in terms of profit maximization. In practice, one should balance these benefits with the added complexity of using different thresholds for different products. In a fully digitalized system with barcodes and QR codes that are scanned prior to products being shipped, we conjecture that the complexity cost is minimal. Also, in practice, shipping rules should be reevaluated on a regular basis, as new products are introduced and/or costs need to be reevaluated. We recommend updating the shipping age thresholds on a quarterly to an annual basis. This can be done by reoptimizing the thresholds using our approach or using some online optimization mechanisms (see, e.g., Keyvanshokooh et al., 2021).

For proper shelf life management, it is essential that the warehouse management and sales systems are capable of tracking inventory by its age not only for items in storage but also for shipments. Upon production, barcode stickers

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are typically placed on the pallet shrink wrap that indicate the production date and a product identifier (see Figure 15 in the Supporting Information Appendix D). Once the pallets are taken apart, age visibility is often lost because barcodes on individual products only include a Universal Product Code (UPC), which is solely a product identifier and does not include the manufacturing date or expiry date. The next generation of barcodes are two-dimensional (2D) barcodes that can capture inventory age information and can, therefore, carry this information all the way to the store level; however, these are not currently widely used due to their high implementation cost. In the absence of 2D barcodes or a database that is not capable of storing sales information by inventory age, small upgrades to the fulfillment system can help with execution. For example, during the picking process, the picker could be required to scan the pallet-level barcode. If the product on the pallet is past its shipping age threshold, the system can prevent the picker from including the inventory in the shipment or the shipment can be flagged as including "too old" items and this information can be later used to analyze compliance with the policy. Also, visual cues can be used to facilitate the execution by categorizing products according to their freshness needs. For example, yellow stickers can be used on pallets for products with flatter profit functions, indicating their greater tolerance to varying freshness levels while red color can be used for products with stringent freshness needs.

We shared our findings with a supply chain expert who managed BevCo's supply chain technology for 7 years focusing on network strategy. He acknowledged that lack of date visibility creates a challenge in monitoring compliance with the shipping age rules. Products which have passed their shipping thresholds are usually identified during periodic inventory counts and are often donated. He told us that the lack of donation records following an inventory count could be evidence of noncompliance with the thresholds. His team would sometimes hear about plants shipping out their short-coded inventory down to warehouses to avoid disposal at their locations (such shipments are governed by BevCo's *ship-to-satellite* rules as explained in Section 2). He added that such behavior would upset downstream warehouse managers because that aged inventory would then "become their problem." Therefore, "bending the rules" is not uncommon, especially with a lack of visibility regarding the age breakdown of inventory. He said our findings would be welcomed by operations managers in the field (at plants and warehouses) as we show that "flushing older inventory down the supply chain" would either not make a difference or, sometimes, would allow BevCo to save money and reduce waste at the same time. He said, "they would feel comfortable bending the rules further." We also discussed that our model does not capture the possible loss of goodwill when consumers buy aged inventory. Typically, such matters are concerns for the quality team at BevCo (which also sets *best taste limit* dates), not the supply chain managers, who primarily care about operational costs. Finally, he found Proposition 2 very interesting and thought it can be helpful for the whole industry as it shows that shipping age thresholds should be driven solely by the trade-off in cost parameters.

5.4 | Adaptation to retailer's problem

In our base model, we solve the optimum maximum shipping age problem in a two-tier centralized supply chain with the manufacturer as the decision-maker who manages the retail store inventory and fully reimburses the retailer for the cost of expiration. However, the problem of setting shipping age thresholds also applies to retailers that distribute products from their own distribution centers and are financially responsible for waste at their retail stores (as is typical of many non-DSD products). Our method can be adapted for this type of retailer.

A U.S.-based retailer with nearly 2000 stores disclosed to us that they will soon start implementing their own maximum shipping age rules for shipment to the stores because shipping aged inventory "adds to payroll" (referring to the substantial labor cost associated with handling), which becomes a problem when many products end up expiring at stores. This policy change is one of the retailer's two planned projects related to food waste management for the coming year.

In our base model, the manufacturer's profit function consists of three components: (i) a gross profit earned on units shipped to retail stores, (ii) the expiration cost at the warehouse, and (iii) the expiration cost at retail stores. For the retailer's problem, the profit function would also include three components: (i) the gross margin earned on units sold to the end consumers, which is equal to the retail price minus the wholesale price minus the handling cost associated with picking items at the warehouse, transporting from the warehouse to the stores, and stocking at the store; (ii) the expiration cost at the distribution center, equal to the wholesale price plus the disposal fee; and (iii) the expiration cost at the retail stores, equal to the wholesale price plus the disposal fee, plus the handling cost associated with moving expired units from the store shelves to the backroom, transporting them to the disposal site, and making stock adjustments in the system.

5.5 | Future research and projects for practitioners

We see several opportunities for future research and initiatives related to shelf life and food waste management.

One is related to the handling of aged and expired items at warehouses. One BevCo warehouse manager disclosed to us that their waste management contractor picks up one truckload of waste from his warehouse weekly, which includes returns from stores. Instead of disposal, because much of this food is still safe for consumption, we recommend partnering with food rescue organizations to schedule donation pick-ups. Alternatively, distributing and marking down the product through discount retail stores could be considered, and future research could identify optimal markdown levels. One

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European retailer we spoke to reported that a 50% markdown is very effective at curbing food waste but hurts profitability, while a 30% markdown is a better way to preserve margins. Future research could investigate the potential cannibalization effect of such markdowns on fresh, full-priced items.

Another potential research opportunity is the study of age-related inventory allocation policies among multiple retail stores. In our analysis, we have assumed that the warehouse inventory is distributed among retailers according to the SOF (a.k.a. FIFO) issuing policy, in a cyclic manner, on a per unit basis; however, there may exist more efficient allocation policies that take into account store characteristics and the current age breakdown of their inventory. In a recent practice-based research project conducted in collaboration with Hershey, Rajan and Wang (2016) segment retail stores into two clusters: fast movers versus slow movers. Warehouse inventory is picked in a FIFO manner for fast movers and in a LIFO manner for slow movers. The researchers optimize the cutoff value that determines fast versus slow movers. While there have been attempts to tackle this problem analytically in simplified settings (Karaesmen et al., 2011), there is still a need for well-performing heuristics for large-scale real-world problems.

Finally, another promising research opportunity relates to incentives for warehouse managers. Establishing maximum shipping age thresholds should motivate warehouse managers to prevent inventory aging through adequate product rotation and better ordering. However, we have anecdotal evidence of a lack of compliance with shipping rules; a BevCo warehouse manager told us that "flushing the old inventory down the supply chain" is better because "some will sell." Indeed, retail-level expiration does not impact his performance metrics while written-off inventory at his warehouse does. Hence, there is a need to better align incentives with the company-wide goal of profit maximization given the relative costs of expiration at the warehouse and retail store levels.

6 | CONCLUDING REMARKS

Shelf life management is an essential part of food waste reduction efforts. The latest comprehensive study on *unsaleables* identifies eight best practices to reduce food waste through shelf life management that are to be carried out collaboratively between retailers and manufacturers (FMI, 2017). The study advises retailers to rotate old inventory on the shelves, conduct audits to make sure rotation is practiced, and mark down near-expired products. At the same time, manufacturers should ship products in a FIFO fashion from their warehouses, carefully and accurately set shelf lives of new products with thorough testing during development, set case sizes that reflect product shelf lives and sales velocity (e.g., a slow-moving item with a short shelf life should be packed in two to four unit cases as opposed to 24), track and record the remaining shelf life as products move through the supply chain, and set minimum shelf life thresholds for shipments to retail stores. While each of these practices is noteworthy subject for future research, in this paper we focused on the last one.

The necessity for manufacturers to establish minimum shelf life rules (or equivalently, maximum shipping age thresholds) has been emphasized in multiple other studies (Raftery, 2011; Wipro, 2010). Although past industry reports provide several general principles (e.g., items nearing the threshold should be put on hold with an action plan), no specific recipe is provided as to how these rules should be set up. Our study, firmly grounded in practice through a real-life dataset, satisfies this need by providing a structured and practical approach to setting maximum shipping age thresholds, and we provide guidelines to implement it in practice. Our analysis shows that carefully optimizing shipping age thresholds can considerably improve on the existing one-size-fits-all rules.

ACKNOWLEDGMENTS

The authors are grateful to Brian Spearman, Paul Hamilton, and Tim Russell for their valuable support and thank seminar participants at the London Business School and Nebraska College of Business for their comments. The authors are also thankful to the department editor Chelliah Sriskandarajah and the anonymous associate editor and reviewers for their feedback on previous versions of this paper.

ENDNOTES

[1] The SOF policy is the recommended inventory issuing policy by industry associations (FMI, 2017) and is commonly required, including at our collaborator. It is often referred to as the *first-in-first-out policy* (FIFO). To differentiate between the inventory issuing policy at the warehouse and the method of inventory depletion at the retail stores, we refer to the former as SOF and latter as FIFO.

[2] In the rest of the paper, we refer to the *ship-to-trade* threshold at BevCo as being 6 weeks because the overwhelming majority of the SKUs in our dataset (450 out of the 455) have a shelf life that is longer than 12 weeks.

[3] https://www.atsinc.com/blog/how-many-pallets-fit-in-trailer-explained

[4] We extrapolate each SKU's expected profit by multiplying it with the actual total sales quantity across all stores in the dataset and dividing this number by the model-generated total order quantity from the five stores in our analysis.

[5] https://www.gs1.org/standards/barcodes/2d

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How to cite this article: Akkas, A., & Honhon, D. (2023). Determining maximum shipping age requirements for shelf life and food waste management. *Production and Operations Management*, *32*, 2173–2188. https://doi.org/10.1111/poms.13963

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